CLICK-A-THON 2026 · ATLYS TRACK

From feature spec to insight — automatically.

An agentic analytics system on ClickHouse that instruments, analyzes, and explains.

3 traced agents | ClickHouse Cloud | LibreChat + Subagents | Vector + OTel | Langfuse + ClickStack

team · till_the_last_row

THE PROBLEM

Every new feature restarts a slow, manual loop

- **Instrument** — someone decides what to track and hand-writes a tracking PRD.
- **Schema** — engineers design & create ClickHouse tables weeks later.
- **Analyze** — analysts write queries, chase segments, build charts.
- **Explain** — by the time insights land, the team has moved on.

Context is lost across every handoff — and a **surprise 6th feature spec** drops in the final hours. Systems tuned to the 5 known specs show their seams.

OUR SOLUTION

Three agents that collapse the loop into one traced run

AGENT 1

Instrumentation

Reads any feature spec, designs optimal ClickHouse DDL (types, ORDER BY, partitioning, MVs), validates + smoke-tests on Cloud, and pushes the schema to GitHub.

AGENT 3

Context Keeper

Maintains a living OKF context bundle. On every new table it bumps the version, logs a changelog diff, and surfaces contradictions in the imperfect base context.

AGENT 2

Analytics

Reads the fresh context, pushes aggregation into ClickHouse (never raw rows), and writes PM-ready insights with the *why* and confidence scores.

Orchestrated by **LibreChat Subagents** — Instrumentation is the parent: **spec → schema → context → insight**, one Langfuse trace end to end.

HOW IT'S BUILT · LIBRECHAT

All three agents run inside LibreChat

Not a bespoke framework — each agent is a configured **LibreChat Agent** (system prompt + skills + tools), chained through native **Subagents**. Agents have no shell: every DB, git, and file action goes through MCP servers.

Subagent chain

Instrumentation is the parent and invokes Context, then Analytics, as isolated subagents — keeping conversation control and combining the result.

MCP tools only

DDL + smoke-tests, read-only SELECTs, git push, and context files are all exposed as MCP servers — safe, auditable tool boundaries.

Conversational UI

The same LibreChat surface is the PM's chat with the Analytics agent for on-demand, ad-hoc questions — every turn traced.

Native tracing

LibreChat's Langfuse integration traces each agent + the full chain; models served as `opus-4.x` via a LiteLLM gateway.

`clickhouse_write_tools` `clickhouse-cloud (read-only)` `clickhouse_git_write` `filesystem`

PROOF · THE UNSEEN 6TH SPEC

Pipeline result: Promo / Coupon at Checkout

Pipeline Result: `unseen_data` (Promo / Coupon at Checkout)

5 insights computed from live ClickHouse data | 5,363 events, 2,100 users | Jun 8–28 2026 | Context v7

Insight	Metric	Key Value	Confidence
I1 — Apply rate is 27.6%, but 31.6% of code entries are rejected	M1: Coupon Apply Rate	580 applied / 2,100 shown	**High (0.90)**
I2 — EXPIRED5 code generates 56% of all rejections — retire it	M2: Reject Mix	149/268 from EXPIRED5; #1 reason: `min_cart_not_met` (30%)	**High (0.85)**
I3 — Coupon users convert at **1.54× the baseline** (63.1% vs 40.9%)	M3: Conversion Lift	Android strongest at 1.77×; iOS weakest at 1.39×	**Medium (0.75)**
I4 — SUMMER20 costs ₹1,055/event at 20% discount; total margin erosion ₹813K	M4: Margin Cost	SUMMER20 > ATLYS15 > FIRST10 > WELCOME > FREESHIP(₹0)	**High (0.85)**
I5 — Android has strongest lift but highest actionable reject rate (34%); Thailand is the most margin-erosive destination (8.7%)	M5: Segment Performance	iOS leads volume; Desktop worst reject rate (40%)	**High (0.80)**

Top action items for PM

1. **Kill or replace EXPIRED5** — it's a dead code generating half of all rejections and frustrating ~150 users
2. **Show min-cart threshold before code entry** — `min_cart_not_met` is the #1 reject reason and entirely preventable with UX
3. **Cap SUMMER20 or convert to flat discount** — at 20% it's the most expensive code and its ₹339K total discount dwarfs others
4. **Fix Desktop coupon field prominence** — 40% reject rate suggests users enter codes speculatively

Autonomous run — 5 insights computed from live ClickHouse (5,363 events · 2,100 users), each with a metric, key value, confidence score, and PM action items. Context v7.

INSIGHT QUALITY · THE "WHY", NOT JUST THE "WHAT"

Insights a PM would act on

1. **M1 (Apply Rate)**: 27.6% of users who see the coupon field successfully apply — with a 40.4% entry rate and 31.6% rejection rate among those who attempt
2. **M2 (Reject Mix)**: EXPIRED5 code drives 56% of all rejections (149/268). Top reason across all codes is `min_cart_not_met` (29.9%), followed by `already_used` (28.0%), `expired` (22.4%), `invalid_code` (19.8%)
3. **M3 (Conversion Lift)**: Coupon users convert at 63.1% vs baseline 40.9% — a **1.54× lift** (+22.2pp). Android shows the strongest lift (67.8% vs 38.3%)
4. **M4 (Margin Cost)**: SUMMER20 is the costliest coupon at ~20% discount rate (₹338,623 total). FREESHIP costs ₹0 in discount. Total coupon discount: ₹813,317 on ₹6,344,375 of couponed cart value (12.8% margin cost)
5. **M5 (Segment Performance)**: iOS leads in volume (874 users) and apply rate (29.1%); Android shows strongest conversion lift. GB and AU show highest apply rates (~31.5%); Turkey trails at 22.6%. Thailand has the highest margin cost per destination at 8.7%.

Apply rate, reject mix, conversion lift, margin cost, and segment performance — cross-cut by device, geo, and destination.

1.54×	56%	₹813K	40%
coupon conversion lift vs baseline	of rejections from one dead code (EXPIRED5)	total margin erosion identified	desktop reject rate — worst segment

CONTEXT FRESHNESS · NO STALE SNAPSHOT

The Context Agent catches what's stale & contradictory

Context contradictions surfaced

- `coupon-reject-designed-not-observed.md` claimed zero `coupon_rejected` rows — **stale**: 268 rows now present in populated data
- M3 computed via base-table self-join per `coupon-conversion-lift-single-table-gap.md` (funnel agg cannot express per-user sequencing)
- K6 (SUMMER20 campaign) confirmed: code is present and active with the highest per-event discount (₹1,055 avg)

On the new table landing, the agent re-checks the base context and flags a claim that is now stale — proving Analytics reasons from just-updated context.

Provable freshness

Versioned Markdown bundle + newest-first changelog — every update is a diff a judge can read.

Compute in ClickHouse

LLM only narrates aggregates — never streams raw rows. Token-safe by design.

No trace, no credit

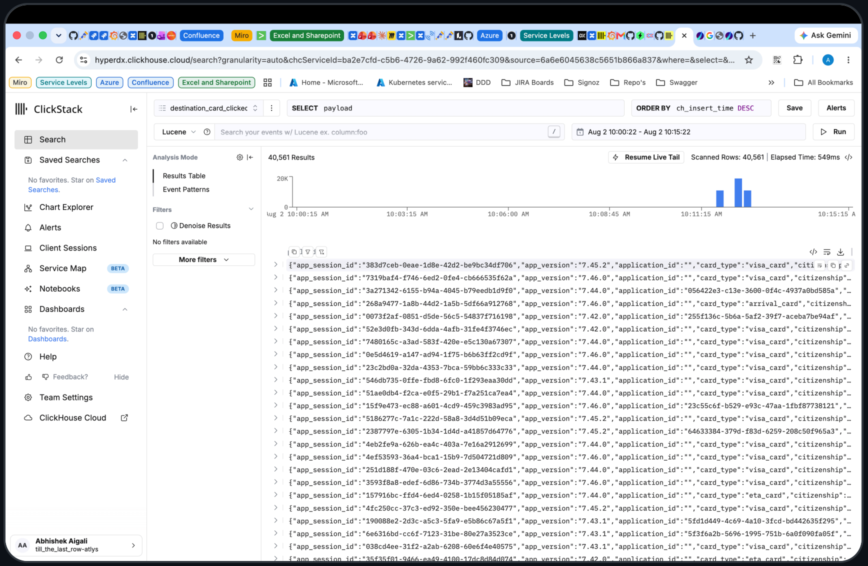
Every agent step is a named Langfuse span; the full chain is one trace.

DATA PIPELINE · INGESTION

Vector + OpenTelemetry Collector → ClickHouse

A containerized **Vector** + **OTel Collector** pipeline transforms events and streams them into ClickHouse Cloud — decoupled from the agent flow.

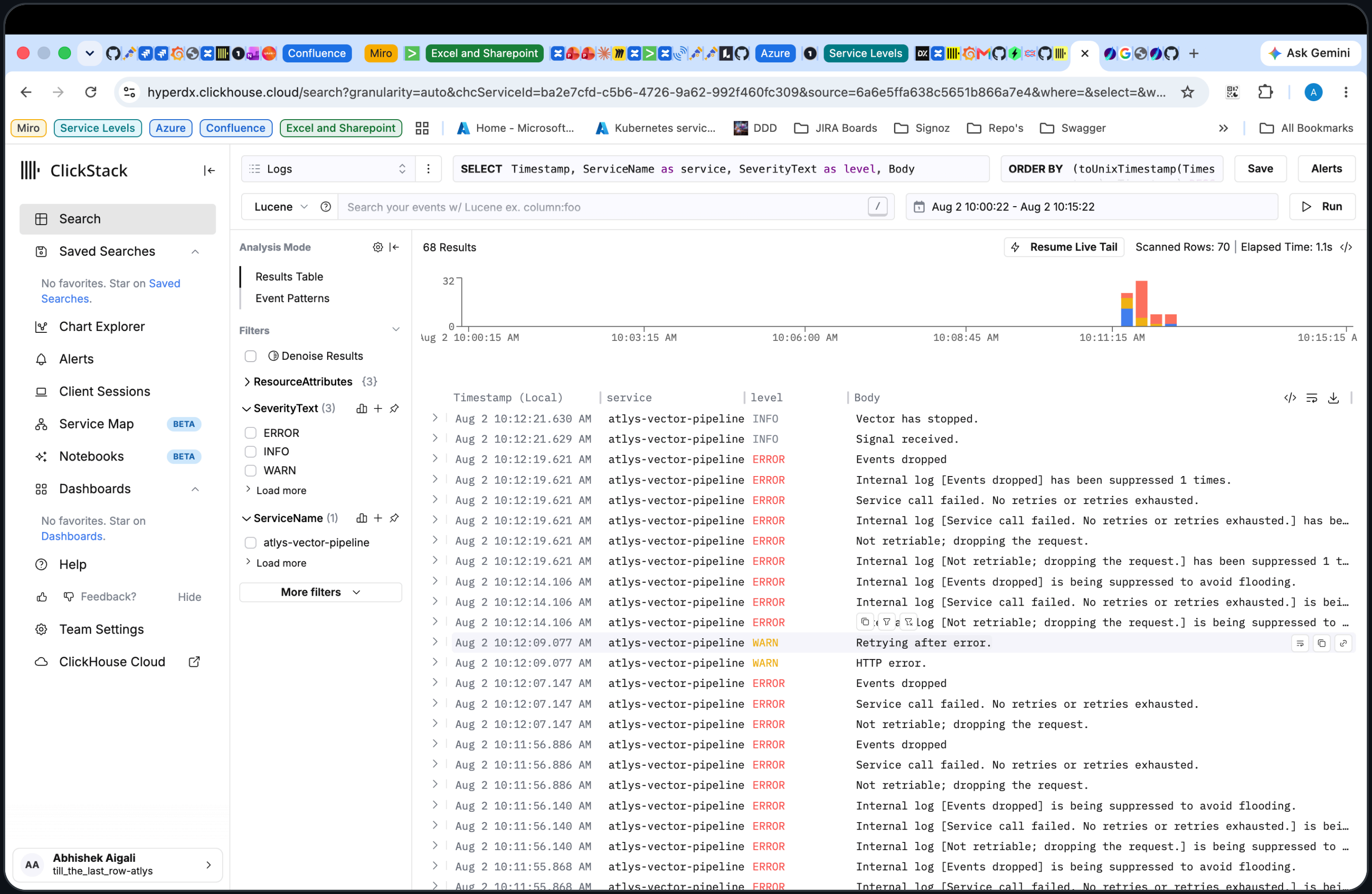
vector-pipeline	-	-	-	0.59%	15 secor	
atlys-otelcol	592c5402200d	otel/opente	24224:24224 Show all ports (4)	0.38%	15 secor	
atlys-vector	e056096ca521	vector-pipe		0.21%	15 secor	



Live services: **atlys-vector** + **atlys-otelcol** feeding ClickHouse — 40,561 events landed and queryable within seconds.

OBSERVABILITY · CLICKSTACK / HYPERDX

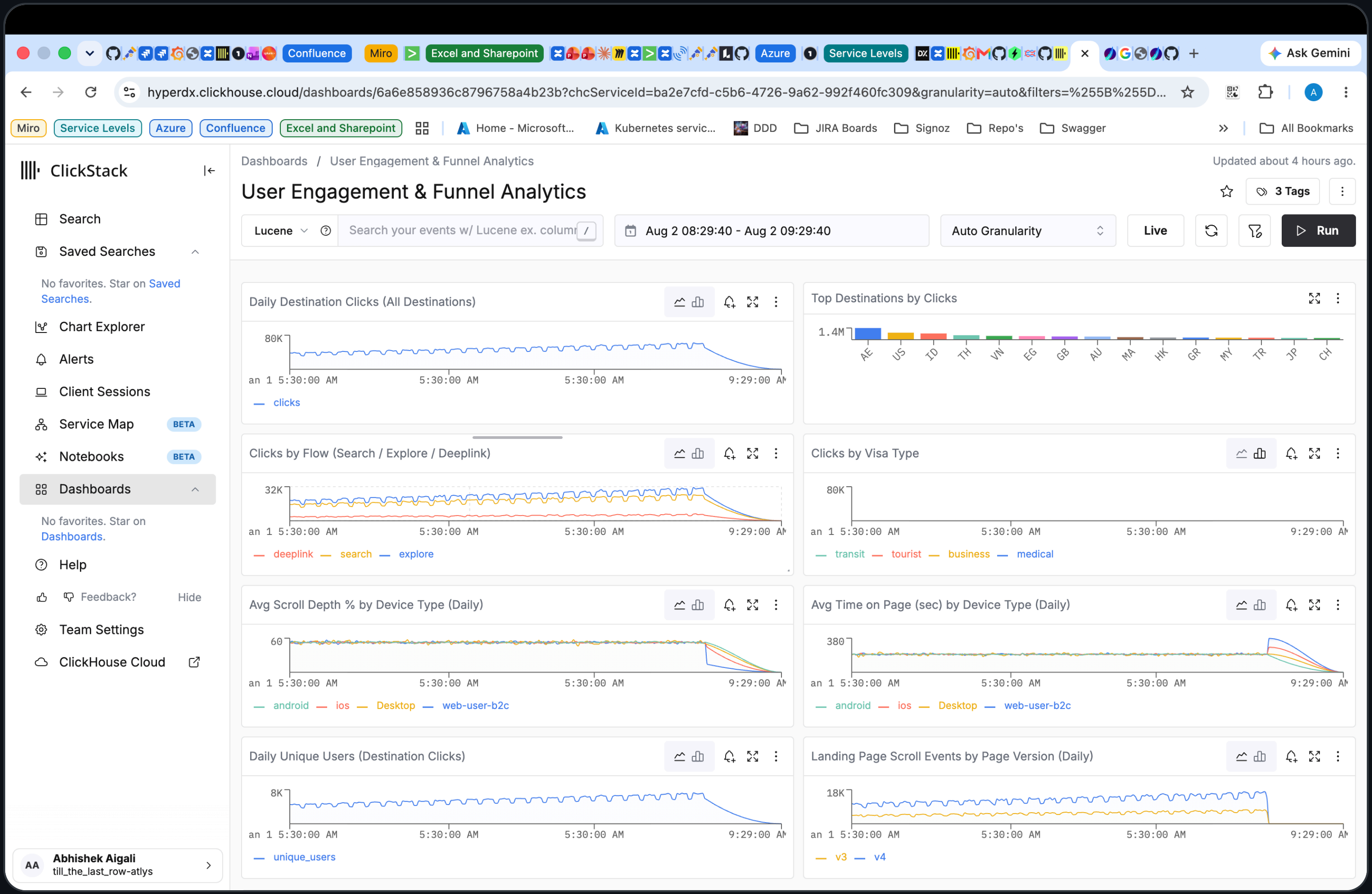
Full platform observability on ClickStack



Pipeline logs, severities, and service health for **atlys-vector-pipeline** in HyperDX / ClickStack — complementing Langfuse (LLM traces) with system-level observability.

VISUALIZATION · ENGAGEMENT & FUNNEL

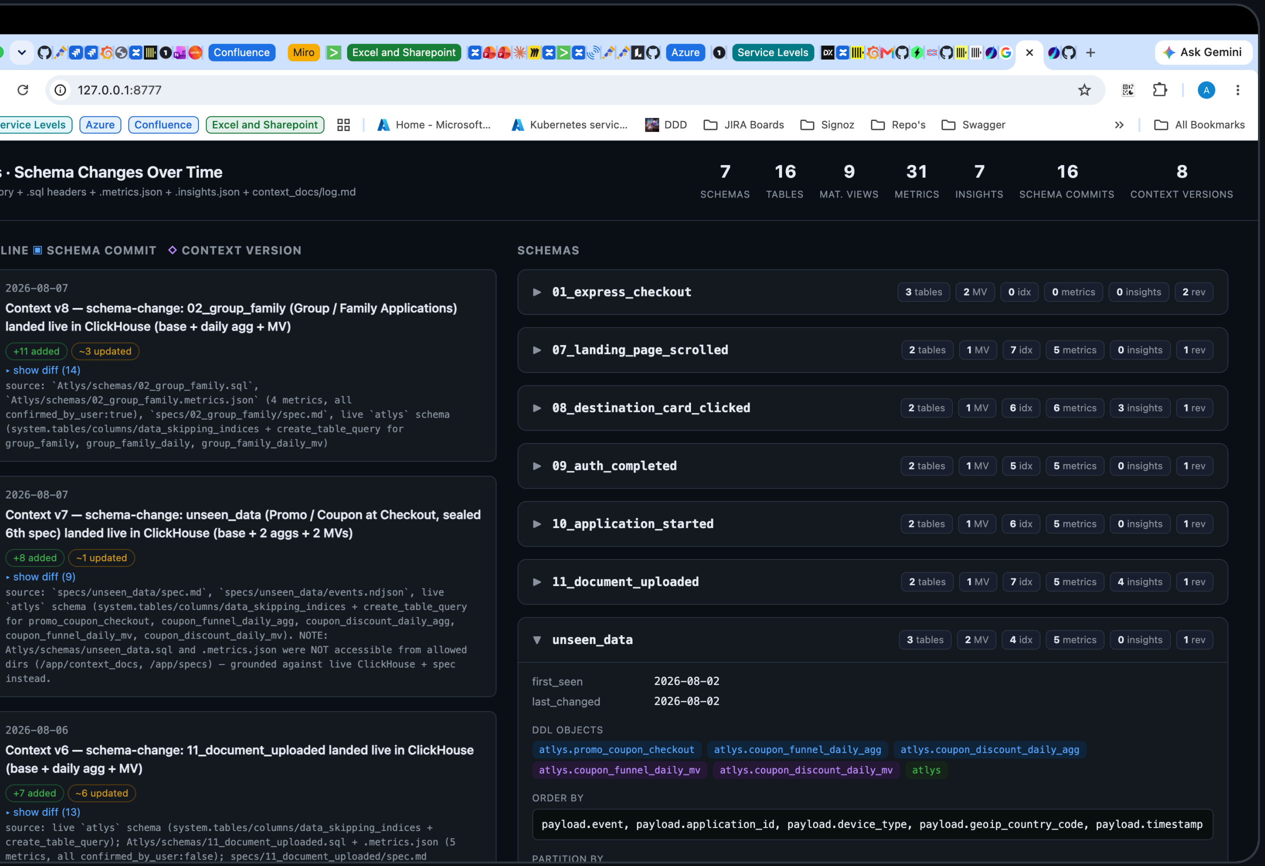
User Engagement & Funnel Analytics dashboard

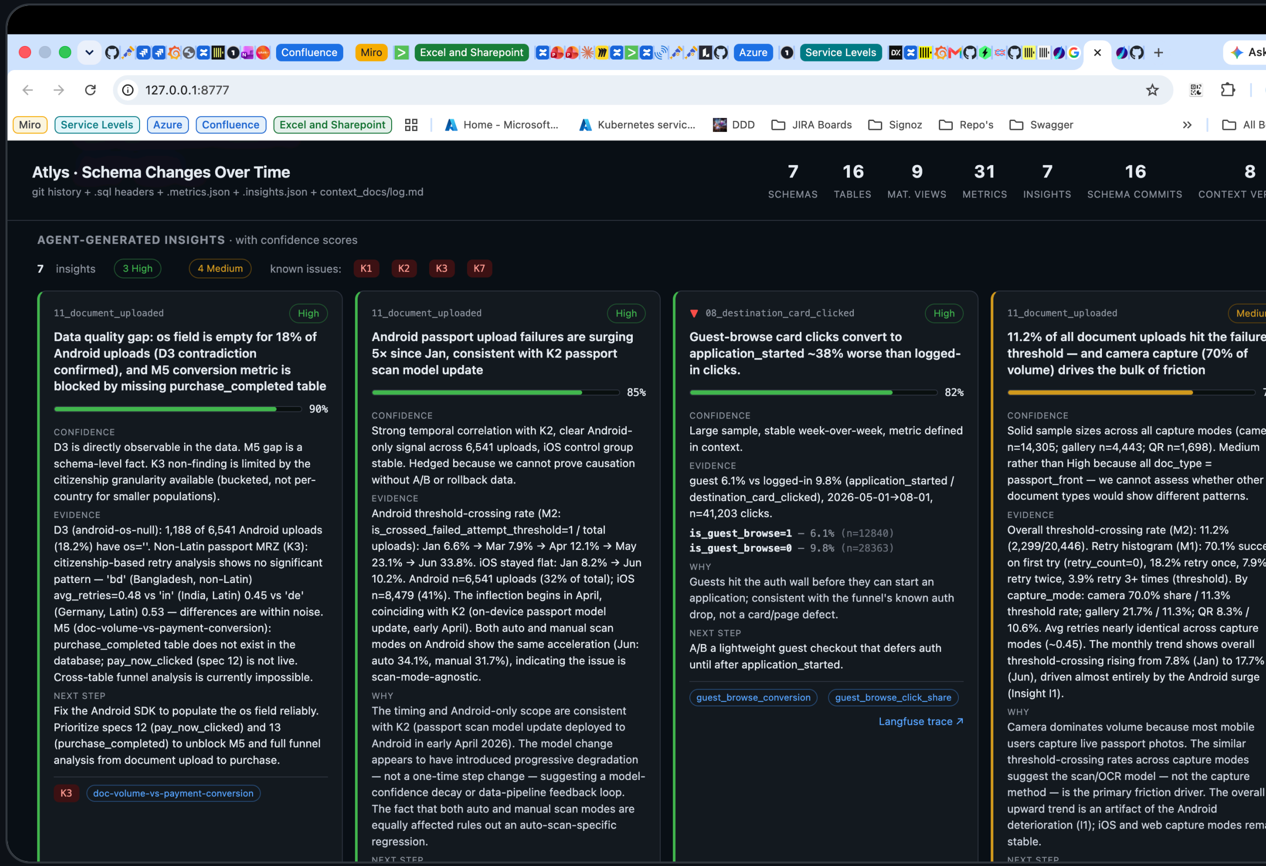


Destination clicks, flow & visa-type breakdowns, scroll depth and time-on-page by device, and daily unique users — all served live from ClickHouse in ClickStack.

VISUALIZATION · SCHEMA EVOLUTION

Schema changes over time — with agent insights





A live view over git history + `.sql` / `.metrics.json` / `.insights.json` + `context_docs/log.md`: **7 schemas · 16 tables · 9 MVs · 16 schema commits · 8 context versions** — each commit tied to a context version, diff, and confidence-scored insights.

WHY WE WIN

Built for the unseen spec — not the five we knew.

- **General by design** — parses free-form specs, no hardcoding for known specs.
- **End-to-end & traced** — spec → schema → context → insight in one Langfuse chain.
- **Full stack** — Vector ingestion, ClickStack observability, live schema & funnel dashboards.
- **Insights, not charts** — confidence-scored actions a PM ships today.